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Passionate programmer with a deep love for building cutting-edge, reliable and simple solutions.

Education

M.S. in CS @ New York University

01/2024 – Exp 12/2025

- GPA: 3.8/4.0

- Courses: Compiler, Parallel Programming, Operating Systems, Big Data App Dev, Multicore CPU Programming.

B.E. in ECE @ Shanghai Jiao Tong University

09/2019 – 08/2023

- Core GPA: 3.7/4.0

Projects

Parallel implementation of hash visualization algorithm

- Programmed with **OpenMP** to implement a parallel version of the hash visualization algorithm used in OpenSSH.
- Achieved a **work efficient** solution that is embarassingly parallel on the pixel rendering level.

Infinite world–The random map generation

- Implemented algorithm based on expression trees that generates Minecraft-like infinite maps using **procedural content generation** (PCG).
- Parallelized the algorithm using **Rake and Compress** algorithm to reduce the span to $\mathcal{O}(\log n)$, where n is the depth of the expression tree, hence accelerated the procedural genration speed.
- Used **Raylib** to achive a Minecraft-like game interface.

Compiler construction

- Built a compiler that compiles from **ML-ish** code down to **RISC-V** in **OCaml**.
- Implemented a **type checker** that detects code illness at compile time.
- Used **Control Graph Analysis** for optimizations including constant propagation, dead-code elimination, and loop optimization.

Work Experience

SuXiang Automobile Tech Ltd. (Startup)

Shanghai, China

Full Stack Dev Intern

08/2023 – 12/2023

- Developed a high-performance **single-page web app** using **Elm** (frontend), **Go** with **Gin** (backend), and **MariaDB**.
- Deployed mail server, Mattermost, and Nextcloud using **Docker**, integrating Authentik for **SSO** management (LDAP, OAuth, SAML).
- Automated **CI/CD** pipeline using Drone with Gitea, improving development and deployment efficiency.

Samoyed Cloud Tech, Ltd.

Shanghai, China

Big Data Infrastructure Intern

06/2022 – 09/2022

- Processed large-scale financial datasets on **HDFS** for distributed storage and computation.
- Developed **Spark**-based ETL pipelines to clean, transform, and aggregate financial data for AI/ML training.

Skills

- **Programming languages:** C++, C, Go, OCaml, Python, Java, SML, Scala, Shell, Elm.
- **Tools & DevOps:** Git, Docker, Kubernetes, Nginx, Cloud (AWS, GCP), Hadoop, Ray.
- **Databases:** MySQL, MariaDB, SQLite, MongoDB, Redis, Hive.